Egg consumption and risk of cardiovascular diseases and diabetes: a meta-analysis.

OBJECTIVES: To assess the dose-response relationship between egg consumption and the risk of cardiovascular diseases (CVD) and diabetes. METHODS: We systematically searched MEDLINE database through December 2012. Fixed- or random-effects model was used to pool the relative risks (RRs) and their 95% confidence intervals (CIs). Subgroup analyses was performed to explore the potential sources of heterogeneity. Weighted linear regression model was used to estimate the dose-response relationship. RESULTS: Fourteen studies involving 320,778 subjects were included. The pooled RRs of the risk of CVD, CVD for separated diabetes patients, and diabetes for the highest vs lowest egg intake were 1.19 (95% CI 1.02-1.38), 1.83 (95% CI 1.42-2.37), 1.68 (95% CI 1.41-2.00), respectively. For each 4/week increment in egg intake, the RRs of the risk for CVD, CVD for separated diabetes patients, diabetes was 1.06 (95% CI 1.03-1.10), 1.40 (95% CI 1.25-1.57), 1.29 (95% CI 1.21-1.37), respectively. Subgroup analyses showed that population in other western countries have increased CVD than ones in USA (RR 2.00, 95% CI 1.14 to 3.51 vs 1.13, 95% CI 0.98 to 1.30, P = 0.02 for subgroup difference). CONCLUSIONS: Our study suggests that there is a dose-response positive association between egg consumption and the risk of CVD and diabetes.